Can a Cheap Text Monitor Tell Right from Wrong Reasoning?

CSI 436/536 Machine Learning (Fall 2026) — Project P Midterm One-Pager Group: [members] Oct 20, 2026

Research question (fixed by the project).

Do pretrained representations improve downstream ML tasks, particularly under limited training data? We turn the yes/no question into a measurement problem. With the class algorithm g and training data fixed, the *only* varied factor is the feature source φ, and we report the gap function

$$\Delta(\theta) \;=\; L_{\text{test}}\big(g \circ \varphi_{\text{simple}};\, \theta\big) \;-\; L_{\text{test}}\big(g \circ \varphi_{\text{pretrained}};\, \theta\big), \qquad \theta = (\text{train size } n,\ \text{distribution shift},\ \text{perturbation}).$$

Deliverable: a measurement report — one summary sentence (yes/no with effect size and CI) plus three boundary curves $\Delta(n)$, Δ(shift), Δ(noise).

Task and data (application carrier).

A cheap *text monitor* for math reasoning: given a problem and one solution step (text only), classify the step as correct or incorrect. **PRM800K** (OpenAI, MIT license): human-labeled step-level correctness on MATH solutions (∼800k labels). **Math-Shepherd** (GSM8K, automatically labeled): transfer test set only. Perturbed variants (rule-based typos, number formatting, LaTeX style): robustness test sets. Labels come with the datasets; no LLM labeling is used.

Method (strictly within the rules).

Pretrained text embeddings — `all-MiniLM-L6-v2`, `bge-base`, `bge-large` — are used *frozen* as feature extractors only (forward pass, no fine-tuning). On top we train only models covered in class: logistic regression (L7) and linear SVM (L11). Control arm: TF-IDF (word / char n-grams) with the same classifiers. Protocol: input = problem + current step; **group split by problem** (no step of one problem crosses train/test, preventing leakage); primary metrics **balanced accuracy / AUROC** (labels are imbalanced; no raw accuracy); 10 stratified re-samples per n, mean ± CI; C selected by grouped k-fold CV inside the training set only; the test set is touched once; all seeds fixed.

Experiments.

E1 learning curves $\Delta(n)$, $n \in \{20, 50, 100, 500, 2000, \text{full}\}$ (sample efficiency); **E3** generalization Δ(shift): train on PRM800K (MATH) → test on Math-Shepherd (GSM8K) — reported as a *compound* shift (problem type, source, and labeling method change together; no single-factor attribution), plus train/test gap; **E4** robustness Δ(noise): AUROC degradation under rule-based perturbations; **E2** three embedding models (novelty axis); **E5/E6** optional (abstention via Chow's rule; a preliminary predictability probe).

Pre-registered predictions (locked Oct 19, before any runs).

(1) gap(n) peaks in the small-sample region ($n \leq 100$) and stays non-negative; (2) Δ(shift) remains positive; (3) embedding degradation under perturbation is no larger than TF-IDF's. Falsified predictions will be reported as boundary findings, not hidden.

Current progress.

Full pipeline implemented and smoke-tested (data prep → features → grouped CV → evaluation → figures; SST-2 bring-up passed). A first run on PRM800K exposed a *label-construction bug on our side*: using only the `chosen` completion per step yielded a 99.8% positive label rate and degenerate metrics. We fixed it to use all human-rated completions and are re-running; all current numbers are preliminary. Encouragingly, the ranking-based signal was unaffected by the bug: at small n the embedding–TF-IDF gap is consistently positive (e.g., +0.03 AUROC at n=50, MiniLM, preliminary).

Challenges and plans.

Class imbalance (stratified sampling + balanced metrics); compound-shift attribution; honest reporting if pre-registration is falsified. Next: finish E1 (top priority) → E3 → E4 → E2 on the corrected dataset; final talk will show $\Delta(n)$, Δ(shift), Δ(noise).